

Limbomorphs

Alex Alvarez¹, Michael Levin^{2,3}

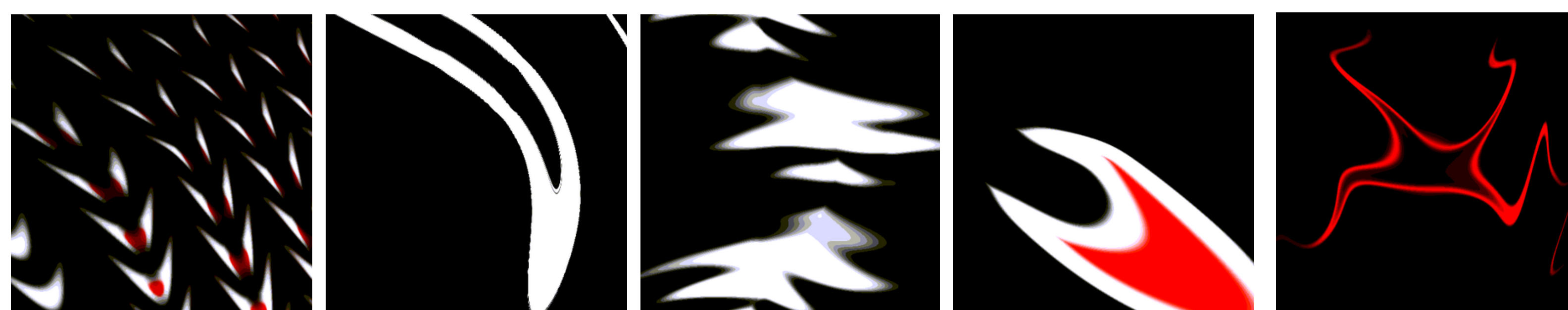
1. Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology

2. Allen Discovery Center, Tufts University

3. Wyss Institute for Biologically Inspired Engineering, Harvard University

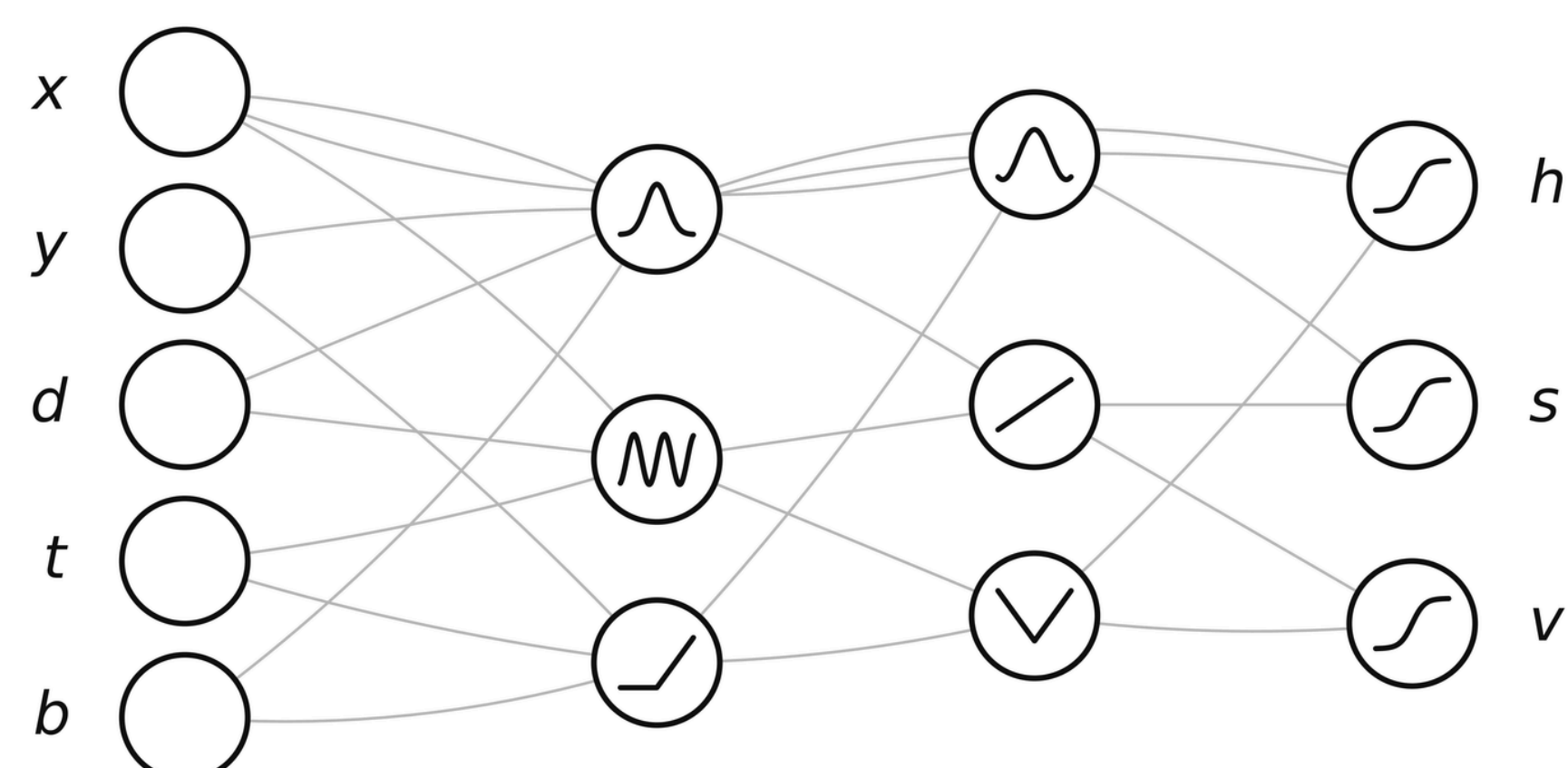
One sentence summary

Lifelike creatures with species-specific reactions to perturbations emerge from Gifbreeder despite the absence of explicit agents, environments, or local interactions.



Gifbreeder

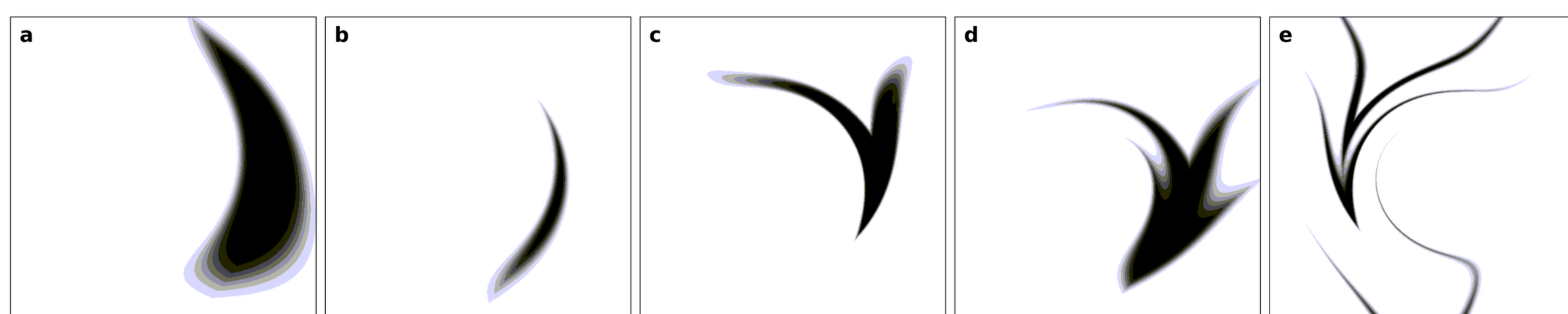
- Gifbreeder is an interactive evolutionary platform inspired by Picbreeder which uses NEAT-based search over CPPN representations to evolve animated GIFs.
- The CPPN's inputs for each pixel are its coordinates, radial distance from center, time, and a bias constant. The outputs are displayed as HSV channels.
- Gifbreeder defines a spacetime field** rather than a simulated world.



$$F_{\theta}(x, y, d, t, b) \rightarrow (h, s, v)$$

Discovery

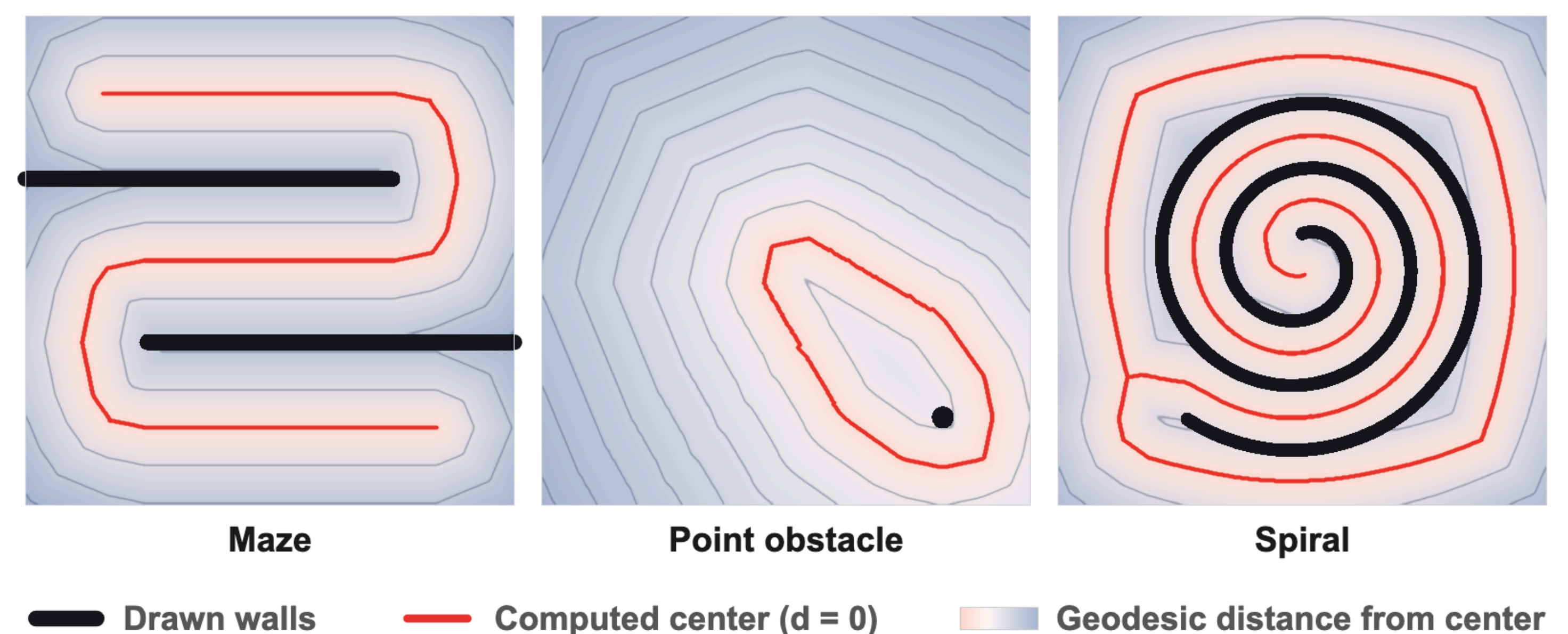
- The emergence of lifelike beings in Gifbreeder was unexpected since it was designed to be an art tool.
- At first, Gifbreeder patterns were desolate and abstract. After several evolutionary runs, **we began to notice expressions that exhibited apparent animacy**.
- Once a limbomorph was found, it could be used as a branch-off point to evolve a family of creatures.



Nine limbomorphs of increasing morphological complexity. They begin as a large undifferentiated mass (a) before becoming thinner (b), developing a forked tail (c), a secondary bifurcation (d), and eventually fine, filamentous appendages (e).

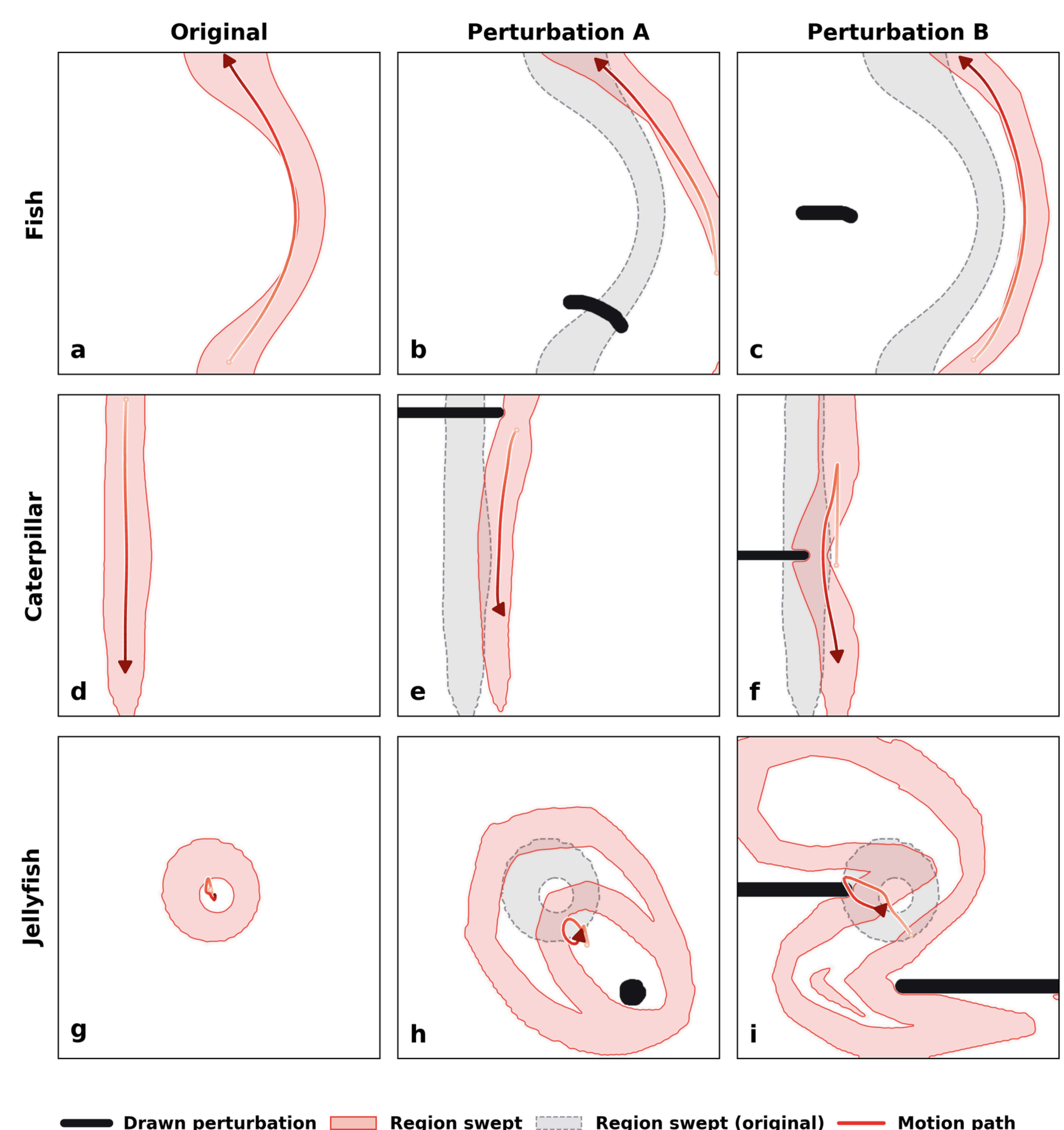
Perturbations

- Limbomorphs don't have local rules or receptors, however, one way to think about interacting with limbomorphs is to **warp the CPPN's inputs**.
- When the user draws with the perturbation tool, distance from center is replaced by the geodesic distance from the computed center ridge.



Results

- Fish** is extremely avoidant to the perturbations(a-c).
- Caterpillar** navigates away (e) or towards (f) the perturbation depending on the region.
- Jellyfish** stretches its topology to envelop the perturbation(h) and even orients its body around the maze(i).



Conclusion

- Species-specific reactions to perturbations suggest that behaviors of spatial competence and goal-directedness may arise in a system with no explicit agent, environment, or interaction rules.
- Could limbomorphs provide a useful model system for investigating how basal cognition can emerge from field dynamics alone?

Abstract & References

